# ELI — Perception

ELI v2.2.9

August 2026

## Contents



## ELI Perception — Vision, Voice, Wake Word, Tone, OS Control

`eli/perception/` (~20 files). ELI's senses and hands: local vision, speech-to-text, text-to-speech, a **self-trained wake-word detector**, a **voice-profile/tone** subsystem, and OS control. All local, no APIs, no third-party accounts.

### Files

| File | LOC | Role |
|---|---|---|
| `audio_stt.py` | ~1.6k | STT + mic capture + ducking + adaptive pause + wake/voice capture |
| `wakeword.py` | ~450 | **self-trained, music-robust wake-word detector** (openWakeWord features + custom head) |
| `voice_profile.py` | ~420 | **prosody + labelled-emotion** (tone/question detection) |

| File | LOC | Role |
|---|---|---|
| `tts_router.py` | 1195 | Piper/pyttsx3/espeak router + `char:`/`clone:`/`natural:` voice resolution; `neural_fallback_state()` makes a neural→Piper fallback observable |
| `voice_fx.py` | 257 | **character voices** (HAL/TARS/Rick/GLaDOS/JARVIS) — base voice + ffmpeg effect chain; user preset store |
| `tts_xtts.py` | 351 | **voice cloning** from a reference sample (Coqui XTTS-v2). Zero-shot: `add_clone()` only registers a reference clip, conditioning happens at synthesis. Bundled in the Linux AppImage from v2.1.65; when absent the voice still registers and synthesis falls back to Piper — **loudly**, not silently, which was a live fault |
| `vision.py` | 692 | |
| `os_controller.py` | 573 | |
| `screen_locator.py` | 410 | locate UI elements on screen |
| `gaze_engine.py` | 358 | |
| `log_rotation.py` | 225 | log housekeeping |
| `analyze_pdfs/image/mesh/csv.py` | ~600 | file-type analysers |
| `ambient_vision.py` | 207 | periodic screen glances (off by default) |
| `local_whisper_stt.py` | 316 | |
| `voice_worker(_streaming).py`, `eli_listen.py`, `extract_equations.py` | small | workers/helpers |

## Vision (`vision.py` + `ambient_vision.py` + `analyze_image.py`)

The working local-vision stack (see memory `eli-image-analysis`): - VL model (default Qwen2.5-VL, configurable) loaded via `Qwen25VLChatHandler`; **CPU clip forced** by monkeypatching `mtmd_cpp.mtmd_init_from_file(use_gp` — the GPU mtmd clip path **segfaults** on compute-7.5 cards. - **Hot-swap**: unload text model → load VL → infer → restore in `finally`, holding `gguf_inference._LLM_CALL_LOCK`. - **Co-resident fast path**: a small model (Moondream2 Q4) loaded first, with the text model's ctx capped (`vision_coresident_text_ctx`) so both fit 8GB; `prefer_fast=True` uses it without a swap. - Images downscaled to `vision_max_image_px` (1280) to avoid context overflow; `repeat_penalty`/`top_k`/`top_p` set to kill a repetition loop. - `ambient_vision.py`: optional periodic screen glances (OFF by default). A guarded daemon re-reads the toggle/interval each cycle and **skips a glance whenever the shared LLM lock is busy** (so it never steals the model mid-reply). Stores a short description as memory for rolling awareness.

## Speech-to-text (`audio_stt.py`, `local_whisper_stt.py`)

A large, feature-dense STT module: - Mic capture + Whisper transcription (`local_whisper_stt`). - **Output ducking** — lowers system sink volume while listening (`_eli_duck_output`/`_eli_restore_output` via `wpctl`) so ELI doesn't hear itself. - **Echo/self-hearing suppression** (`_eli_echo_like_assistant_output`, `_eli_media_probably_audible`). - Transcript cleanup: `_cleanup`, `_collapse_repeated_phrase`, `_eli_fast_command_alias`

(maps spoken phrases to commands), `_is_safe_direct`/`_allow_direct_chat` (gates which utterances bypass to chat). - ALSA stderr suppression to keep the console clean. - **Duration-adaptive end-of-phrase pause** (`_listen_adaptive_pause`, 2026-06-08) — stock `sr.listen()` reads `pause_threshold` once per phrase, so one value can't be both snappy for commands and tolerant of long dictation. A faithful copy of sr's capture with a single dynamic condition: short commands finalise after `ELI_STT_SHORT_PAUSE` (0.5s) of silence; a prompt speaking past `ELI_STT_LONG_AFTER` (12s) needs `ELI_STT_LONG_PAUSE` (2s), so a mid-sentence pause no longer cuts it. Flag-gated (`ELI_STT_ADAPTIVE_PAUSE`) with fallback to stock `listen()`. - **Generic mic-capture script** — one mechanism (no second microphone) drives both wake-word enrollment and voice/emotion training: a list of `{prompt, sink}` steps, each captured (past the TTS-echo guards) clip routed to its sink, the next cue spoken, a `done` callback after the last. `begin_capture_script` / `begin_wake_enrollment` / `begin_voice_training`. - **Acoustic wake hook** — when unarmed and a wake model is trained, the captured audio is scored by `wakeword` and, if it fires, the wake word is injected so the existing `VoiceGate` arms — catching the wake word even when whisper transcribed the music. **Per-turn tone hook** — on a real command, `voice_profile.classify_tone` is run and published on a side-channel for cognition (below).

## Wake word (`wakeword.py`, 2026-06-08) — self-trained, robust over music

Transcription-based wake matching can't hear "computer" over loud music (whisper transcribes the music). Instead ELI **trains its own** detector, 100% locally: - **Positives**: the wake phrase synthesised by ELI's OWN **Piper TTS** across several voices and speeds. - **Augmentation (the robustness)**: each positive is mixed with noise/music at random SNRs, so the classifier learns to spot the wake word *through* a music bed. - **Features**: openWakeWord's open, bundled melspectrogram→embedding extractor (no account, no download barrier). - **Custom head**: a small torch classifier ELI owns; `train_model()` → `models/wakeword/eli_wake_head.pt` (gitignored). `WakeDetector.score_audio`/`is_wake` slide a 1.5s window. Validated: clean wake = 1.00, wake+loud music (3 dB) = 1.00, hard-negative/music-only = 0.00. - **User-settable phrase** — `get/set_wake_phrases` persists any phrase ("change the wake word to athena"); it feeds both this model and the transcription matcher. - **Personalisation** — enrolled real-mic clips of the user are folded in as heavily- weighted positives. - Actions: `WAKE_SET` / `WAKE_TRAIN` / `WAKE_ENROLL`. Fully fallback-safe (`ELI_WAKE_ACOUSTIC=0`; no model → the transcription matcher).

## Voice profile + tone (`voice_profile.py`, 2026-06-08) — foundation for emotion

Deliberately SEPARATE from the wake word — this is *how* the user speaks, the basis for tone/emotion and question-vs-statement: - **Prosody (real, numpy only)**: per-clip autocorrelation **F0/pitch** track, energy, voiced ratio, speaking rate, and a **terminal-pitch slope** (`analyze_prosody`). - **Question vs statement** (working): rising terminal pitch ⇒ question (`question_or_statement`). - **Labelled emotion**: `add_labelled_sample` + a nearest-centroid classifier (`train_emotion_classifier`) over neutral/happy/angry/excited/sad — robust for the small data a quick enrollment yields, upgradeable to a NN without touching callers. - **`classify_tone`** returns emotion + confidence, arousal, and the question/statement cue. `build_profile` learns the user's baseline so tone is scored *relative* to how they normally sound. - **`TRAIN_VOICE`** runs one unified guided session (wake reps + the same line said in each emotion); modules stay separate, the user does one flow. - **Wired into cognition**: STT publishes the per-turn read on a fresh, timestamped side-channel (`set_last_tone`); `engine._build_enhanced_system` reads `get_last_tone` and prepends a speaker-voice cue so ELI adapts its delivery (warmer if upset, brisk if energetic). `ELI_VOICE_TONE=0` disables. Complementary to the text-based `cognition/tone_analyzer` (persona preferences), not a duplicate.

## Tone / emotion adaptor (`cognition/tone_adaptor.py` + `cognition/emotion_palette.py`)

Decides the emotion ELI **expresses** and drives three output channels at once.

- **Two identifiers** (the "2 systems"): *acoustic* (`voice_profile.get_last_tone` — the user's voice this turn) and *semantic* (`detect_text_emotion` — the content of what they said). `_fuse()` combines them (agreement boosts confidence; voice slightly favoured).

- **Empathetic response policy**, not mimicry: a distressed user → `tender`, heat → `calm`, playful → `playful`. An explicit override (`set_tone("be comedic"/"talk street")` via `SET_TONE`/`CLEAR_TONE`) wins outright and persists.
- **Extensible palette** (`emotion_palette.py`): 20 built-in tones (comedic, professional, deadpan, street_smart, gremlin, tender, …), each carrying a persona directive, Piper prosody profile, and avatar expression; users add their own in `config/voices/tones.json`.
- **Three output channels**: text (a directive folded into `_build_enhanced_system`), voice (`tts_router._piper_prosody` overlays the tone's pace/expressiveness/pauses **and `_apply_tone_pitch` shifts the actual audio pitch** so sad sounds lower, ecstatic higher — applied on the Piper, natural-neural and clone paths), and face (`world/avatar/persona_mapper` + the animated `gui/widgets/eli_face.py` take the expression).
- **Core personality is never overwritten** — the directive explicitly shades *how* ELI delivers, not *who* he is (emergent-voice principle). `ELI_TONE_ADAPT=0` disables.

## Animated face liveness (`gui/widgets/eli_face.py` + `cognition/expression_state.py`)

The procedural face reacts in real time to what ELI is *doing*, via three cheap flags in `expression_state` (priority: thinking > speaking > tone): - **Lip-sync** — the mouth animates while TTS plays (`tts_router._run_tts` sets the `speaking` flag; amplitude-driven when available, else a natural talk oscillation). - **Thinking look** — a focused/reflective face while the model generates (the GUI sets `thinking` around `is_generating`). - Otherwise it shows the current expressed tone. Pure-Qt, embedded in the world panel.

## Emotional memory + proactive check-in (`cognition/emotion_timeline.py`, v2.1.31)

`tone_adaptor` shades the *current* reply and then forgets: the acoustic read lives in a 12-second slot, the semantic read is recomputed from the last utterance. That makes ELI **reactive**. The emotion timeline is the durable record that makes it **proactive**.

- **Persistence** — every fused read (neutral included, so the baseline isn't built only from bad turns) is appended to `emotion_events` in `user.sqlite3`, with the utterance that produced it, the arousal, and **the action ELI ran on the preceding turn**.
- **`assess()`** returns EVIDENCE, never a phrase: the sustained run and its dominant emotion, the `neutral→negative` transition, the ELI action at the turn the mood turned, and whether it is *unusual for this user* — measured against their own baseline, so a naturally reserved person is never read as upset.
- **Gates** (all must pass before ELI raises anything): a run of N reads rather than a single spike, per-read confidence above a floor, a credible baseline, and a cooldown so ELI cannot nag. `assess()` reports its `reason` either way, so a *non*-check-in is explainable too.
- **Two surfacing paths.** In-conversation, `evidence_block()` is folded into the system prompt and **the model decides whether and how to raise it** — the wording is ELI's, never a template (see [ELI No Hard-Coded Responses]). Unprompted, the proactive daemon runs the same assessment on its tick, LLM-synthesises an opener from the evidence, and pushes an `emotion_checkin` onto the suggestion queue (+ a pending proposal so a plain "yes" routes). With no resident model the daemon emits **nothing** rather than a canned line.
- **`trend_line()`** runs every turn regardless, giving ELI continuity on how the conversation has been feeling.
- **User control** — Settings ▸ Cognition ▸ *Emotional awareness*: master toggle, exchanges before a mood counts, minimum confidence, check-in cooldown, recent-mood window, baseline history. All read at call time (no restart), all defaulting to the shipped constants. A *Tone of voice* dropdown in the same tab pins the register ELI answers in (Auto + the 20 palette tones) via `tone_adaptor.set_tone`. Kill switch: `ELI_EMOTION_MEMORY=0`.

## Text-to-speech (`tts_router.py`)

Multi-backend router with graceful fallback: **XTTS-v2 natural neural** (opt-in) → **Piper** (packaged binary under `tts_piper/` or `models/tts/piper`) → **pyttsx3** → **espeak-ng / espeak**. Resolves voices from several candidate dirs so a packaged build or a source checkout both work.

*Natural neural voice (`natural:<id>`, `tts_xtts.py`).* XTTS-v2 driven by its **built-in studio speakers — no clone/reference needed** — is the human-like, expressive, punctuation-aware path. `list_natural_voices()`/`synthesize_natural_wav()` are cheap to probe (never load the ~1.8GB model just to enumerate) and **fall back to Piper** when the neural extra isn't installed, so a `natural:` voice never goes silent. It is selectable at the **startup model picker / first-boot wizard** ("Voice engine: Piper vs Natural") — the choice is persisted *before* hardware tuning so its VRAM is a deliberate up-front decision (`ELI_XTTS_DEVICE` pins the device); low-VRAM boxes keep fast Piper. Enable with `pip install -e ".[natural]"` (from the ELI project root).

*Expressive Piper prosody.* Piper at bare defaults sounds flat and clips full stops. The router now passes tuned `--sentence_silence` / `--length_scale` / `--noise_scale` / `--noise_w` (`_piper_prosody_args()`, env-overridable, `ELI_TTS_PROSODY=0` to disable) so `?/!/.` land as intonation and pauses — the "Amy sounds flat" fix.

*No robotic voices.* `list_voices()` hides the espeak `sys:` voices and Piper `low`/`x_low` tiers by default (they sound synthetic), surfacing them only as a genuine last resort when the box has nothing better, or when `ELI_TTS_ALLOW_ROBOTIC=1`.

**Voice library** (`runtime/voice_assets.py`) — the shipped pack is a starting point, not the limit. The module fetches the upstream Piper index (**166 voices across 45 languages, 38 of them English** — US/GB, Scottish, Northern and Southern English) and exposes it through `list_available_voices()` with per-voice presence, download size and licence, cached on disk so the picker still works offline. `download_voice()` streams from the index's exact paths and **verifies the published md5** before the file is put in place, so a truncated or proxy-mangled download is rejected rather than surfacing later as a corrupt-model crash. The GUI path is Settings > VOICE / TTS > *Get more voices / accents…* (`gui/widgets/voice_downloader.py`); every voice dropdown repopulates on install.

*Redistribution policy.* `RESTRICTED_VOICE_NAMES` (ryan — CC-BY-NC-SA; lessac — Blizzard/Lessac, not cleared; cori — pending) are never put in a release asset. The rule is keyed on the **voice name, so every quality variant is covered**, and `scripts/asset_release_policy.py` imports it so the build and the runtime cannot drift. The user may still download them for personal use; the dialog states this rather than hiding the voice.

*Config integrity.* A `.onnx` without its `.onnx.json` cannot be loaded by Piper, so it is invisible to `list_voices()` while still occupying ~60 MB. `incomplete_voices()` finds these and `repair_voice_configs()` fetches just the missing ~5 KB config. `tests/test_voice_library.py` guards against shipping the broken state again.

**Character voices** (`voice_fx.py`) are a base voice + an ffmpeg effect chain (pitch / speed / filters) — HAL, TARS, Rick, GLaDOS, JARVIS built in, and users create their own (`char:<name>`, preset store at `config/voices/characters.json`). Because the ideal bases for HAL/TARS/Rick (lessac/joe/ryan) are restricted or unbundled, each preset carries an **ordered, gender-matched `fallback` chain**; `resolve_base_voice()` walks base → fallbacks → default against the *installed* set, so a character can never fall through to an unusable or foreign-language voice (it previously landed on a Czech voice and garbled the speech). Download the ideal base and it is used automatically. **Voice cloning** (`tts_xtts.py`, `clone:<name>`) reproduces a voice from a short reference clip via Coqui XTTS-v2 — an opt-in extra (`eli-v2.0[clone]`) that falls back to a normal voice when the model isn't installed rather than going silent. Both `char:` and `clone:` resolve through `synthesize_wav` (browser voice) and `_run_tts` (host playback), and appear in the GUI voice picker.

## OS control (`os_controller.py`)

**Platform-aware** (Linux/Windows/macOS) screenshot, volume, keyboard, clipboard. `take_screenshot(region)` picks the right tool per platform (`gnome-screenshot`, PIL `ImageGrab`, platform CLIs); fails gracefully where a capability isn't available (e.g. area screenshots on some Windows installs). `screen_locator.py` finds UI targets for click/automation.

## File analysers

`analyze_pdfs`, `analyze_image`, `analyze_csv`, `analyze_mesh`, `extract_equations` — typed handlers behind the `ANALYZE_*` actions, feeding extracted content (text/OCR/structure) into the grounded pipeline.

## Honest assessment

- **Strong:** this is the layer that makes ELI genuinely multimodal and local — eyes (VL + OCR + ambient), ears (Whisper + ducking), mouth (Piper/espeak), hands (cross-platform OS control). The vision co-residence + hot-swap + CPU-clip workaround is real engineering against hard 8GB-GPU constraints.
- **Weak / watch:**
  1. **Vision latency** — the hot-swap unloads/reloads the text model per glance; even the co-resident path runs CPU clip (~3.5s). Acceptable, not fast.
  2. `audio_stt.py` is a **1.48k-line** module mixing capture, ducking, echo suppression, command aliasing, and cleanup — wants splitting.
  3. **Residual "7B" comment** in `ambient_vision.py` ("the 7B vision model hot-swaps…") — cosmetic, but inconsistent with the model-agnostic line; should read "the vision model". (Flagged for a future scrub.)
  4. `gaze_engine.py` is experimental and off by default — fine, but it's carried weight.
  5. Platform tools are detected at call time; a machine missing `gnome-screenshot`/`wpctl`/`piper` degrades silently to fallbacks (good) but there's no single "what perception capabilities do I actually have here?" probe surfaced to the user.

---

## Update — 2026-06-09 (VRAM-aware STT; drag-drop keeps the full path)

- **STT is VRAM-aware** (`local_whisper_stt`). faster-whisper defaults to GPU only when the card is big enough for whisper AND a typical main model (≥12 GB total VRAM), else CPU — so on an 8 GB card whisper no longer preloads ~2 GB and starves the main model (observed: free_vram 4083 MB → gpu_layers=11; after the fix free_vram 7092 MB → gpu_layers=99). `small.en` int8 on CPU is ~1–2 s for a short command. Explicit `ELI_WHISPER_DEVICE` still wins; `ELI_WHISPER_GPU_MIN_MB` tunes the threshold.
- **Drag-and-drop keeps the FULL path.** A file dropped into the chat input now expands to `[File: <full path>]` followed by the inlined content (was content + basename only), so "fix/examine/edit this file" can resolve the file on disk while "summarise this" still has the content. PDFs keep their path too; combined with the FIX_FILE last-file memory, a bare follow-up "fix it" recovers the file named a turn earlier.